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Origin, evolution and taxonomy of American *Roccella* (Roccellaceae, Ascomycetes)

Abstract The phylogeny and taxonomy of four American *Roccella* species are presented: *R. bajasurensis* (newly described here), *R. decipiens*, *R. gracilis* and *R. verruculosa*. A key to all nine American *Roccella* species is also presented. The data from four molecular markers, RPB2, nuLSU, ITS 1 and 2 and an anonymous locus showed that the specimens of what was believed to be the one fertile species, *Roccella decipiens*, actually belonged to two distinct species: one usually corticolous, facultatively fertile-sorediate species, *Roccella gracilis*, and one strictly saxicolous, obligately fertile species *R. decipiens*. The American *Roccella* species were found to belong to a significantly supported, monophyletic group with most Macaronesian species as its sister group and all other *Roccella* species distributed in the Old World. Vicariance contra long distance dispersal is discussed in explaining biogeographical patterns.

Key words lichenised fungi, biogeography, species pairs, *Roccella bajasurensis*

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Introduction

The genus *Roccella* DC. contains nine species on the North and South American continents. Four of these species are treated in this work: *Roccella bajasurensis* (newly described here), *R. decipiens*, *R. gracilis* and *R. verruculosa*. The other five species, the *Roccella galapagoensis* aggregate, have been treated recently in a separate publication (Tehler *et al.*, 2009).

Roccella is a subtropical genus of lichen fungi with a preference for areas with Mediterranean climates extending into adjacent temperate and tropical regions, and it is predominantly distributed over the northern hemisphere (Tehler *et al.*, 2004; Tehler & Irestedt, 2007). All species are restricted to coastal habitats and may locally be very common. *Roccella* uses *Trentepohlia* algae as a photobiont and the lichens are fruticose with a white-greyish to greyish-brown usually pendent thallus of flattened or terete branches, normally between 5–15 cm long; the cortex is composed of anticlinally arranged hyphae and the medulla is normally white to pale brownish; several species have a characteristically yellow medulla in the holdfast, e.g. *R. phycopsis* and *R. gracilis*. The fruiting bodies are apothecioid, circular in outline, immersed or sessile, up to 2.5 mm in diam., with a pruinose disc. The thalline margin is more or less developed; all species have a characteristically carbonaceous hypothecium, distinct from the neighbouring white medulla. The asci are thick-walled, fissitunicate and contain eight fusiform, 3-septate, hyaline spores, typically within the size range of 15–35 × 4–8 μm. The conidia are filiform and curved; most species produce soredia for vegetative dispersal. Secondary metabolites are typically orcinol and β-orcinol depsides, and aliphatic acids.

The main American distribution is along the Pacific Coast of Upper and Lower California, Peru, the Galapagos Islands and various localities around the Caribbean Sea. On the North American Pacific Coast, it extends to just north of Los Angeles in California (*c.* 34° north). On the Pacific Coast of South America, it extends just south of the equator (south of Lima in Peru, *c.* 14° south). There are no *Roccella* species present in South America south of the Atacama Desert. Three species from North America were treated by Tehler (2002) in the *Lichen Flora of the Greater Sonoran Desert Region*: (i) *Roccella decipiens*, a species that was considered strictly fertile, (ii) *R. peruensis* [recently named *R. gracilis* (Tehler, 2006)], a species that was considered strictly sterile and sorediate, and (iii) a species that was tentatively identified as *Roccella portentosa*. Regarding the two former species, previously there were numerous names in circulation (Esslinger & Egan, 1995) and there was confusion as to what names should be applied to the North American species. The two most commonly, but incorrectly used names, were *Roccella fimbriata* for fertile specimens and *R. babingtonii* or *R. peruensis* for sorediate specimens.

Roccella portentosa, recently recombined to *Roccellina portentosa* (Tehler & Irestedt, 2007), is a strictly South American species which is not found north of the Atacama Desert in Chile. The identification of the Baja Californian specimens as *R. portentosa* was considered as provisional by Tehler (2002) and with this study it has become evident that they belong to a new species (described below).

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Here, we will revise the taxonomy and nomenclature of the remaining *Roccella* species from the Americas that were not treated by Tehler *et al.* (2009), by combining data from molecular phylogeny with morphological evidence. We will also contribute to the growing knowledge on the phylogeny of *Roccella*, and finally discuss the biogeography of this exciting group, in a phylogenetic context.

Materials and methods

Terminal taxa and taxon sampling

The study is based primarily on collections made by A. Tehler from the Galapagos Islands (2005); Peru (1981, 2006); Bonaire, Netherlands Antilles (2006); California, USA (1976, 1993–1996); Baja California, Mexico (1993–1996, 2008); and on specimens from the following herbaria: B, BM, COLO, FH, G, H, HBG, L, M, PC, S, TFMC, TUR, UPS, W.

The primary species sampled for DNA-sequencing in this study were: *Roccella bajasurensis* (newly described here), *R. decipiens*, *R. gracilis*, *R. verruculosa* (Table 1, which is available as 'Supplementary data' on Cambridge Journals Online: http://www.journals.cup.org/abstract_ S1477200009990065). Furthermore, representatives of the other American species, *Roccella albida*, *R. galapagoensis*, *R. lirellina*, *R. margaritifera*, *R. nigerrima*, as well as most of the Old World species, *Roccella allorgei*, *R. babingtonii*, *R. balfourii*, *R. boryi*, *R. elisabethae*, *R. fuciformis*, *R. maderensis*, *R. montagnei*, *R. phycopsis*, *R. sinensis*, *R. tinctoria* (including the former *R. canariensis*) and *R. tuberculata*, were included in the analysis for outgroup purposes. We tried to achieve a sample series with as large as possible morphological variation and with a large and representative geographical distribution. Appointed voucher specimens are deposited in (S).

Molecular techniques and alignment

Four loci have been sequenced and used in the phylogenetic analyses, the nuclear ribosomal internal transcribed spacers ITS 1 and ITS 2 (and the intermediate 5.8s region), the nuclear large subunit ribosomal RNA gene (LSU), the second largest RNA polymerase subunit (RPB2), and an anonymous locus. Extractions, amplifications, and sequencing procedures follow Tehler and Irestedt (2007) and Tehler *et al.* (2009). The only exception is that a newly designed primer, Cal-int3F (ACTTTGCACGAGAGGCCAAGCATT), was used for amplifications and sequencing of the anonymous locus for *Dirina ceratoniae* (the reversal primer, Cal-int1R, and the thermocycling program was unchanged).

The sequence fragments were assembled to complete sequences with SeqMan II(tm) (DNASTAR Inc.). Ambiguous nucleotide positions were coded with the appropriate IUPAC codes. GenBank accession numbers are given in Table 1, which is available as 'Supplementary data' on Cambridge Journals Online: http://www.journals.cup.org/abstract_ S1477200009990065). Alignments were carried out using the computer program ClustalX 2.0.9 (Larkin *et al.*, 2007) under Multiple Alignment Mode with all parameters set to default. No manual post adjustments were made except for cropping sequences in the beginning and the end of the alignment block.

Choice of outgroup

The outgroup and rooting taxon, *Dirina ceratoniae*, was chosen on the basis of findings in earlier analyses dealing with phylogenetic relationships in the order Arthoniales (Tehler & Irestedt, 2007) in which the crustose genus *Dirina* was significantly supported as the sister group to the fruticose genus *Roccella*. To break up the long branch between *Dirina ceratoniae* and the ingroup, numerous *Roccella* species outside the American continents were taken from previous studies and included in the analyses.

Phylogenetic analysis

Parsimony analyses and Parsimony jackknifing frequencies were performed and calculated with the program TNT (Goloboff *et al.*, 2008). Resampling tree searches were done under the New tech search with sectorial search, ratcheting, drifting, tree fusing and driven search options in effect, all using default settings; 1000 replicates submitted to TBR branch swapping were conducted.

Chemistry

For routine screening of secondary chemistry, 'spot-tests' on thalli and apothecia were performed with a water solution of potassium hydroxide ('K') and with a commercial solution of potassium hypochlorite ('C'). High-performance thin-layer chromatography (HPTLC) following Arup *et al.* (1993) was conducted on selected specimens to identify the secondary metabolites present. In addition, thin-layer chromatography (TLC) of acetone extracts was performed in solvent systems A, B, C and G on silica gel 60 F254 layer glass plates of 20 × 20 cm following Orange *et al.* (2001) for specimens of *Roccella bajasurensis*. *Cetraria islandica* was used as a control species for the identification of the fatty acids.

Taxonomy

Roccella DC.

de Lamarck & de Candolle, (1805: 334). Type: *Roccella fuciformis* (L.) DC.

For generic description the reader is referred to Tehler *et al.* (2009).

Key to *Roccella* species from America

1 Medulla in holdfast yellowish.....2
1 Medulla in holdfast never yellowish.....5

2 Thallus branches terete, C-.....***Roccella bajasurensis***
2 Thallus branches flattened or strap like, C+ red.....3

3 Soralia present.....***Roccella gracilis***
3 Apothecia present.....4

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4 Saxicolous, ascospores 5–6.5 μm broad, thallus stiff .....***Roccella decipiens***
4 Corticolous, ascospores 3–4 μm broad, thallus softer.....***Roccella gracilis***

5 Apothecia present, soralia absent or rare.....6
5 Soralia present, apothecia absent or rare.....9

6 Apothecial disc C- (endemic to Galapagos)..... .....****Roccella galapagoensis***
6 Apothecial disc C+ red.....7

7 Apothecial disc faintly C+ red; Medulla dirty-white to brownish, coalescent; thallus branches relatively thicker (fertile form rare, endemic to Galapagos) .....****Roccella margaritifera***
7 Apothecial disc conspicuously C+ red; medulla whitish, nearly byssoid; thallus branches relatively thinner.....8

8 Thallus whitish, branch tips K-, (rare, endemic to Galapagos).....****Roccella albida***
8 Thallus brownish, branch tips slightly K+ reddish .....***Roccella verruculosa***

9 Thallus C-, (endemic to Galapagos)..... .....****Roccella nigerrima***
9 Thallus C+ red.....10

10 Medulla whitish, loose, more or less byssoid; thallus whitish with thin branches; (rare, endemic to Galapagos).....****Roccella albida***
10 Medulla dirty white to brownish, compact hyphae; thallus with thick branches; common.....11

11 C+ red reaction prominent, usually present across the whole thallus (endemic to Galapagos)..... .....****Roccella margaritifera***
11 C+ red reaction weak, usually present only in the uppermost branches (Peru only).....****Roccella lirellina***

*Treated in Tehler *et al.* (2009).

The species

(For representative specimens examined refer to Table 1, which is available as 'Supplementary data' on Cambridge Journals Online: http://www.journals.cup.org/abstract_S1477200009990065)

Roccella bajasurensis Tehler, sp. nov. (Fig. 1)
Type: Mexico. Baja California Sur, Playa San Pedro *c.* 10 km S of Todos Santos, on rocky slope just S of the beach, alt. 50–300 m., 16 Oct. 2007, *Anders Tehler* 9188 (S-F113351, holotype; UPS, isotype).

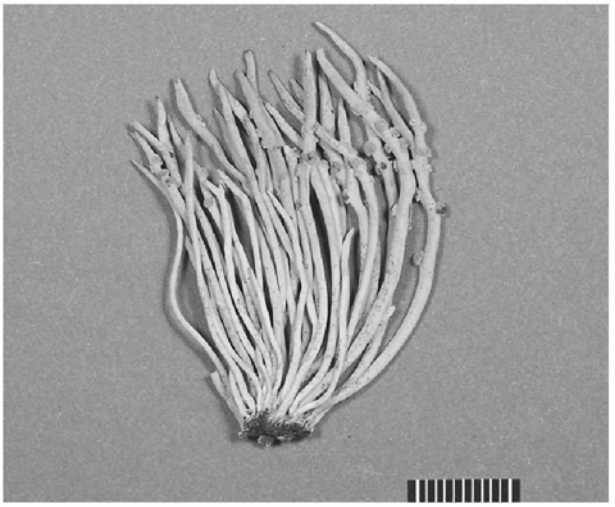

Figure 1 Holotype and DNA voucher of *Roccella bajasurensis*. Mexico, Baja California Sur, Todos Santos, Playa San Pedro, 2008, *A. Tehler* 9188 (S-F113351). Scale mm. Photo Ramona Ubral Hedenberg.

Thallus fruticosus, albido-cinereus; rami numerosi, teretes, 5–10 cm longi. Cortex hyphis verticalibus. Ascocarpia numerosa. Soralia non visa. Medulla albida autem basaliter lutea vel fulva prope substratum. Thallus et ascocarpia C et K non reagens. Thallus acidum protolichesterinicum et acidum lichesterinicum continens.

Thallus fruticose, erect or pendent, branches terete rarely somewhat flattened, 5–10 cm long, white greyish, smooth; soredia absent; medulla white but usually yellowish-brown in the holdfast zone. Ascomata present, sessile; ascospores 23–27 × 5–6 μm. Spot tests: cortex K-, C-, KC-, P-; medulla K-, C-, KC-, P-; Secondary metabolites: protolichesterinic and lichesterinic acids.

Specimens examined. Mexico. Baja California Sur, Punta Lobos beach S of Todos Santos. 23°28′N 106°50′W, 1993, Anders Tehler 7155 (S-L9191); Coastal hillside above Playa Punta Lobos, 2 km SW of Todos Santos (near Rte. 19), on acidic rock, *c.* 40 m. 23°27′N 110°15′W, 1991, T H Nash 29740 (S, UPS); Punta Lobos beach S of Todos Santos. On the small peak, the point, just south of the beach. Alt. 0–50 m, on rock, 2007, Anders Tehler 9171 (S-F113348); Punta Lobos beach S of Todos Santos. On the ridge and mountain behind the point just south of the beach. Alt. 50–300 m, 2007, Anders Tehler 9179 (S-F113350); Playa San Pedro *c.* 10 km S of Todos Santos, on rocky slope just S of the beach, 2007, Anders Tehler 9188 (S-F113351); La Palma *c.* 20 km NW of Cabo San Lucas, near mile 105 on highway 19, 2007, Anders Tehler 9192 (S-F113352); Migriño *c.* 5 km N of the village, *c.* 26 km NW of Cabo San Lucas, near mile 94 on highway 19, 2007, Anders Tehler 9197 (S-F113353); Punta Gaspareno *c.* 50 km N of Cabo San Lucas, near mile 74 on highway 19, 2007, Anders Tehler 9199 (S-F113354); Playa San Pedro *c.* 10 km S of Todos Santos, on rocky slope just S of the beach, 2007, Anders Tehler 9188 (UPS).

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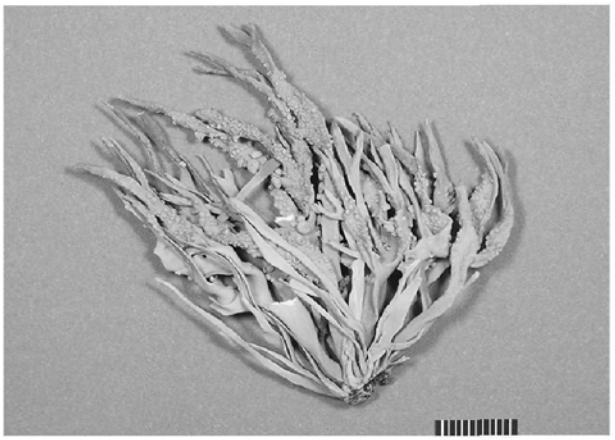

Figure 2 *Roccella decipiens*. Mexico, Baja California, Ensenada, Punta Banda, 2008, A. Tehler 9080 (S-F114301). Scale mm. Photo Ramona Ubral Hedenberg.

Etymology. The specific name *bajasurensis* is latinisation of the short name 'Baja Sur' used for the Mexican State of Baja California Sur.

Distribution and habitat. *Roccella bajasurensis* is only known from Todos Santos south to Cab San Lucas in Baja California Sur, Mexico. It grows on cliffs and rocks near the sea.

Remarks. *Roccella bajasurensis* is the only species in the genus that produces protolichesterinic and lichesterinic acids. It furthermore lacks both erythrin and lecanoric acid, which are otherwise very commonly produced by species of *Roccella*. *Roccella bajasurensis* is thus chemically easy to distinguish from the closely related *R. decipiens* and *R. gracilis* by its C− thallus reaction. The two latter have a conspicuous C+ red reaction. Morphologically *R. bajasurensis* can be separated from these two species by its terete and smooth branches, in contrast to the flat branches in *R. decipiens* and *R. gracilis*. *Roccella bajasurensis* is morphologically more similar to the more distantly related *R. verruculosa* and *R. galapagoensis* but it differs from these by the holdfast, which has a yellow medulla, and by the constantly C negative thallus reaction. *Roccella galapagoensis* may have a either C− or C+ red thallus but the medulla is never yellow in the holdfast.

Roccella bajasurensis was provisionally referred by Tehler (2002) to the superficially similar *R. portentosa* (= *Roccellina portentosa*), a strictly South American, Chilean species.

Roccella decipiens Darb. (Fig. 2)

Ber. Deutsch. Bot. Ges. **15**: 5 (1897), nom. nud.; *Biblioth. Bot.* **9**: 42 (1898), nom. valid. Type: USA, California, San Diego. 1875, *Palmer* s.n. (G, lectotype selected by Tehler *et al.* 2004).

Roccella fimbriata Darb., *Proc. Calif. Acad. Sci.* **21**: 287 (1935). Type: (not found, not seen).

Thallus erect or pendent, main branches distinctly flattened, usually 5–20 cm long, greyish to brown-greyish, usually smooth but sometimes uneven with ridges and wrinkles; soredia absent; medulla white but usually yellowish-brown in the holdfast zone. Ascomata present, sessile; ascospores 16–24 × 5–6, 5 μm. Spot tests: cortex K+ more or less yellowish red (sometimes inconspicuous), C+ red, KC+ red, P−; medulla K−, C−, KC−, P−. Secondary metabolites erythrin and lecanoric acid.

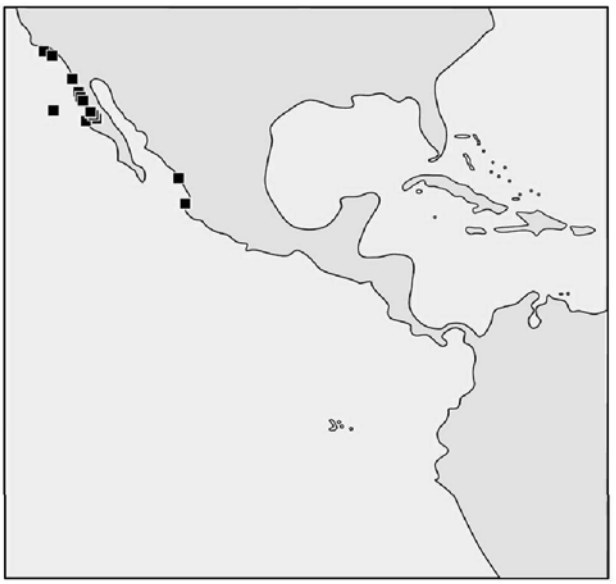

Figure 3 World distribution of *Roccella decipiens*.

Distribution and habitat (Fig. 3). *Roccella decipiens* has its main distribution along the Pacific coast from the Channel Islands off the coast of Los Angeles, California, south to the Vizcainia peninsula including Guadalupe Islands in Baja California, Mexico with a few outpost localities on the Mexican mainland in Sinaloa, Sonora and Jalisco. It grows strictly on rocks in coastal regions.

Remarks. *Roccella decipiens* is characterised by the fertile, non-sorediate thallus. The ascospores are broader (5 – 6,5 μm) then those of *R. gracilis* and the thallus feels stiff and robust compared to the softer consistency in the thallus of *R. gracilis*. The name *R. fimbriata* (Darbishire, 1935) has been frequently utilised for this taxon in California and Mexico. However, *Roccella decipiens* (Darbishire, 1898) is an older name and thus take precedence over *R. fimbriata*.

Roccella gracilis Bory (Fig. 4a, b)

Dict. Class. Hist. Nat. **14**: 631 (1828). Type: Antilles, St Dominque. 1826, *Turpin* s.n. (PC, lectotype designated by Tehler 2006; isolectotype in G).

Roccella ramalinoides Bory, *Dict. Class. Hist. Nat.* **14**: 631 (1828). – Type: French West Indies, Guadeloupe. 'de la Guadeloupe, Antilles'. s.ann., *Jussieu* s.n. (PC, Lectotype [designated by Tehler (2006)].

Roccella montagnei var. *peruensis* Kremp.., *Verh. Zool.-Bot.Ges.* **26**: 442 (1877). *Roccella peruensis* (Kremp.) Darb., *Bull. Herb. Boissier* **5**: 763 (1897). Type: Peru, Lima, 1868–71, *Barranca* s.n. (M-0024563, lectotype designated by Tehler 2002;

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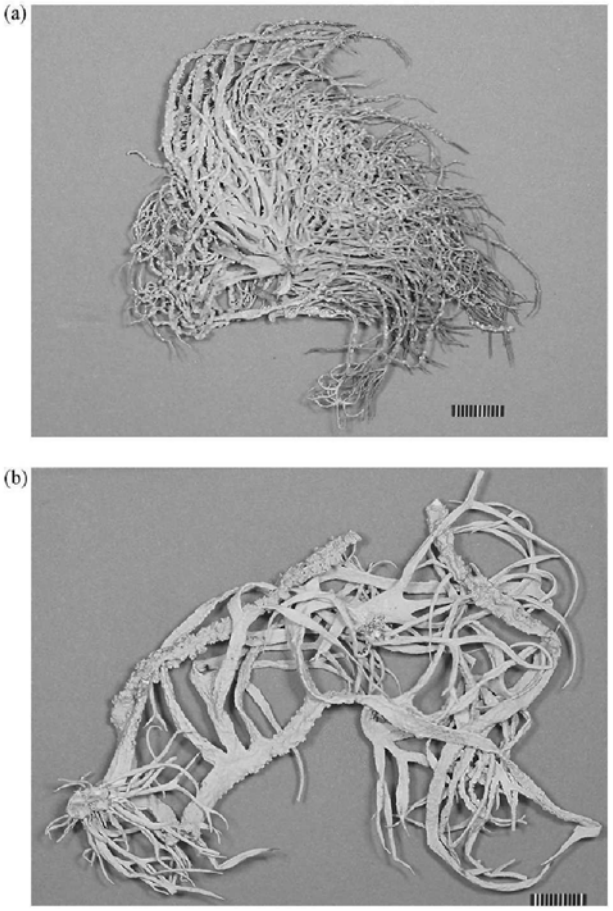

Figure 4 *Roccella gracilis*. Scale mm. (a) sorediate specimen, Mexico, Baja California, Laguna Manuela, 2008, A. Tehler 9126 (S-F114039). Photo Ramona Ubral Hedenberg. (b) fertile specimen, Mexico, Baja California Sur, Todos Santos, Punta Lobos beach, 2008, A. Tehler 9170 (S-F114053). Photo Ramona Ubral Hedenberg.

isolectotypes in BM-000661011, BM-000661013, G, M-0024614, M-0024615, M-0024616, UPS).

Roccella fruticosa Leunis & Frank, *Syn. Pflanzenk.* **3**: 239 (1886). Type: 'Die Californische Orseilleflechten, ...' (Type: not found, not seen).

Roccella difficilis Darb., *Biblioth. Bot.* **45**: 49 (1898). *Roccella humboldtiana* Follmann, *J. Hattori Bot. Lab.* **90**: 262 (2001) [by holotype designation, see Tehler, 2006]. Type: Peru, Payta, 1836, *Gaudichaud* 4 (PC, lectotype designated by Tehler 2002; isolectotypes: B-128641, G, PC).

Roccella mexicana Vain., *Dansk Bot. Ark.* **4**: 24 (1926). Type: Mexico, Puerto de St. Cruz, *Liebman, Lich. Mex.* 7701 (TUR-34668, lectotype designated by Tehler 2002; isolectotypes in TUR-34669 and TUR-34670).

Nomen dubium: *Roccella tinctoria* f. *minor* Kremp. in *Verh. Zool.-Bot. Ges. Wien* **18**: 311 (1868). Type: Mexico, Los Baños, s. ann. *Heller* 155 (PC, lectotype designated by Tehler 2002; isolectotype in G). This collection is the only one AT has seen from the Americas that does not have the typical characters for *Roccella gracilis*, the C+ red soralia and the yellow medulla in holdfast. We presume labels mix up and that the specimens of *Roccella tinctoria* f. *minor* were actually collected somewhere in the Old World and belong to the *R. montagnei* species complex.

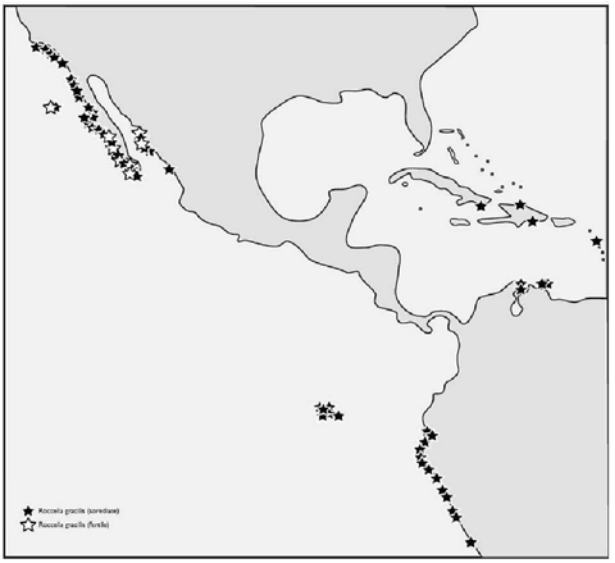

Figure 5 World distribution of *Roccella gracilis*. Filled stars indicate sorediate populations, unfilled stars indicate fertile populations.

Thallus pendent or erect, main branches distinctly flattened, usually 5–15 cm long, creamy to brownish, greyish, usually smooth but often uneven with ridges and wrinkles; soredia commonly present but not together with ascomata, C+ red, in maculiform soralia which may often fuse to cover much of the lobe surface; medulla white but yellowish in the holdfast zone. Ascomata present (Baja California Sur only) but not together with soredia, sessile; ascospores 21–28 × 3–4 μm. Spot tests cortex K+ more or less yellowish red (sometimes inconspicuous), C+ red, KC+ red, P−; medulla K−, C−, KC−, P−. Secondary metabolites erythrin and lecanoric acid.

Distribution and habitat (Fig. 5). *Roccella gracilis* is found along the coasts north from the San Francisco area in California down to Arequipa south of Lima, Peru, including the Galapagos Islands and much of the Caribbean region. Populations may grow to considerable size and richness in suitable localities along the coasts of Baja California and the Galapagos Islands. It grows in coastal regions on any suitable substrate, preferably on the trunks, branches and twigs of various trees and shrubs but commonly also on vertical rocks and cliffs.

Remarks. *Roccella gracilis* is characterised by its usually sterile, sorediate thallus with flat branches and C+ red soredia. Fertile populations are present in the southern part of Baja California Sur including Guadalupe Islands and the adjacent mainland in Sinaloa. Intermediate specimens that contain both soredia and apothecia on the same thallus are extremely rare. *Roccella gracilis* has more slender ascospores (3–4 μm broad) than those of *R. decipiens* and *R. gracilis* has a softer and less robust thallus than that of its sister species *R. decipiens*.

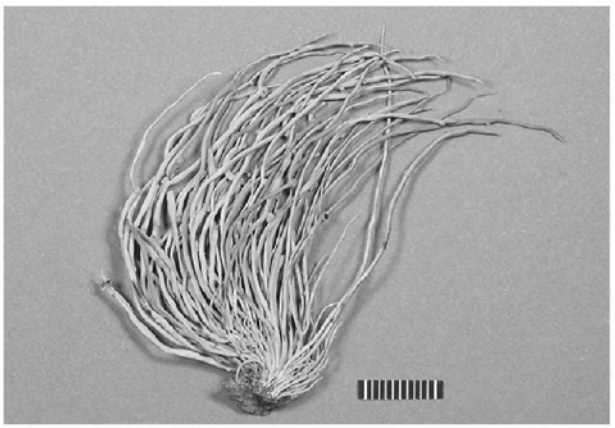

Figure 6 *Roccella verruculosa*. Netherland Antilles, Bonaire, 2008, A. Tehler 8950 (S-F69357). Scale mm. Photo Ramona Ubral Hedenberg.

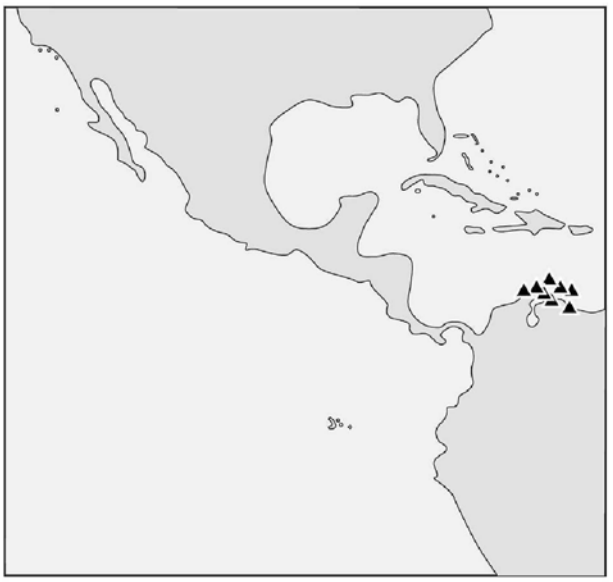

Figure 7 World distribution of *Roccella verruculosa*.

The name *Roccella babingtonii* Mont. has been frequently, but incorrectly used for the sorediate morph of *Roccella gracilis* in California and Mexico. Montagne (1852) described *Roccella babingtonii* as a fertile taxon, soredia was never mentioned by him and it was described on a specimen stated from South America. However, that collection is most certainly mislabelled since it contains a specimen of a species restricted to Kenya and Tanzania in East Africa (Tehler *et al.* in prep.).

Roccella verruculosa Follmann (Fig. 6)

Nova Hedwigia **32**: 22 (1980). Type: Venezuela, Mar Caribe, Islas los Monjes, Monje Grande del Sur, ad saxa granitica. 1973, *Aristeguieta* 8054 (holotype, B-128716; isotypes in B-128715, and in VEN-8054 (according to Follmann (1980) not seen).

Thallus usually pendent sometimes erect, main branches terete rarely somewhat flattened, smooth or foveate, usually 5–20 cm long, brownish to brownish-grey; usually smooth but often uneven with ridges and wrinkles; soredia absent; medulla white, never yellowish in the holdfast zone. Ascomata present, sessile; ascospores 20–28 × 5–6 μm. Spot tests: thallus K−, C+ red, KC+ red, P−; medulla K−, C−, KC−, P−. Secondary metabolites erythrin, gyrophoric acid and lecanoric acid (Follmann, 1980; confirmed by voucher specimen).

Distribution and habitat (Fig. 7). *Roccella verruculosa* is known only from the north-western coastal region of Venezuela and the Netherlands Antilles, Aruba, Bonaire and Curaçao. It grows in coastal regions on vertical rocks and cliffs.

Remarks. *Roccella verruculosa* is morphologically similar to *R. galapagoensis* and *R. albida* and differs from these two species only by its combination of a brownish C+ red thallus with a white to ochre compactly byssoid medulla. *Roccella verruculosa* has usually terete branches but it may also develop thalli with quite flat branches. Such individuals are confusingly similar to *R. decipiens* but *R. verruculosa* is easily distinguished by the total lack of yellowish medulla hyphae in the holdfast. *Roccella verruculosa* is also similar to *R. bajasurensis* but the thallus of *R. verruculosa* is always C+ red whereas *R. bajasurensis* is always C−.

Results

The studied loci and alignments

Eighty-nine ingroup and outgroup sequences from four loci, RPB2, nuLSU rDNA, ITS rDNA, and 88 from the anonymous loci were used for this study. Newly produced were: 59 sequences from the ITS 1 and 2 regions including 5.8S rDNA; 59 sequences from the nuLSU, 59 sequences from the RPB2 and 73 sequences from the anonymous locus; altogether 250 new sequences. We did not succeed to sequence the sample *Roccella fuciformis* 8171 for the anonymous locus (Table 1, which is available as ' data' on Cambridge Journals Online: http://www.journals.cup.org/abstract_ S1477200009990065).

The aligned data set contained 89 terminals and 3066 aligned sites. The sequences obtained from the anonymous locus were compared with calmodulin sequences of *Sclerotinia sclerotiorum* and *Verticillium dahliae* published on GenBank by Carbone and Kohn (1999). A BLAST search was also executed, but no obvious matches could be found. We thus consider this unidentified locus as anonymous, although a potential relationship to the calmodulin gene cannot be excluded. Taken into account that short fragments in the 5′ or 3′ ends of the studied loci are missing for a few taxa (primarily in the anonymous locus and ITS), the length of the sequences and sequence divergences varied as follows. The LSU loci varied in length between 886–896 bp and the sequence divergence ranged between 0.0% to 6.3%; the ITS1 and ITS2 region

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varied in length between 582–636 bp and the sequence divergence ranged between 0.0% to 25.9%; the anonymous locus varied in length between 427–466 bp and the sequence divergence ranged between 0.0% to 33.6%, and the sequence divergence for the 923 bp long region from the coding RPB2 gene ranged between 0.0% to 14.0%. The concatenated alignment consists of 2977 bp. There was no significant incongruence between the individual data sets as evident from the resulting trees (see below).

Phylogenetic analyses

Nineteen *Roccella* species and one *Dirina* species were used in phylogenetic analyses (Table 1, which is available as 'Supplementary data' on Cambridge Journals Online: http://www.journals.cup.org/abstract_ S1477200009990065), with altogether 355 sequences from the four loci. All the American *Roccella* species, *Roccella albida*, *R. bajasurensis*, *R. decipiens*, *R. galapagoensis*, *R. gracilis*, *R. lirellina*, *R. margaritifera*, *R. nigerrima*, *R. verruculosa*, were recovered as monophyletic with significant support. The Macaronesian species, *R. elisabethae*, *R. phycopsis*, *R. tuberculata* and *R. tinctoria* [*R. tinctoria* include *R. canariensis*, see (Tehler *et al.*, 2009)] was placed as sister group (Fig. 8). The individual gene trees (not shown) of both the strict consensus trees and the parsimony jackknife trees were mutually supportive with no statistically significant conflicts.

Parsimony jackknifing (PJ), *combined data set*: The American *Roccella* samples analysed in this study (excluding the five species of the monophyletic Galapagos group, which were previously treated by Tehler *et al.* 2009), were contained in four major terminal groups here interpreted as species based on evidence from both phylogeny [phylogenetic species] and morphology [morphological species]: (1) five samples of *Roccella verruculosa*, PJ frequency 100% containing sequences from two localities; (2) six samples of *Roccella bajasurensis*, PJ frequency 100% containing sequences from six localities; (3) four samples of *Roccella decipiens*, PJ frequency 100% containing sequences from four localities; (4) 39 samples of *Roccella gracilis*, PJ frequency 100% containing sequences from 28 localities in Galapagos, Peru, the Netherland Antilles, and Baja California. This is a further application of phylogenetic species recognition (concordance of multiple gene trees to identify the smallest congruent monophyletic groups between which recombination, as expressed by incongruence between single-gene trees, normally do not occur) in lichenised fungi (Grube & Kroken, 2000; Taylor *et al.*, 2000). In lichen fungal phylogeny, this concept has been successfully applied in several recent case-studies, including the *Stictis/Conotrema* populorum species group, where different species were shown to be able to live either as saprotrophs or as lichens, depending on the substrate (Wedin *et al.*, 2004) and the mineral-accumulating taxa in the morphologically extremely variable *Acarospora smaragdula* group (Wedin *et al.*, 2009).

Parsimony analyses, combined data set: The strict consensus tree (not shown) from the 57 most parsimonious trees 1831 steps long was of the same general topology as that of the PJ-tree (Fig. 8) and the identical four terminal species groups as those in the PJ-tree were recovered. However, in the consensus tree the basal trichotomy of the American species group found in the PJ-tree placing was resolved and *Roccella verruculosa* was recovered as the sister group to the rest of the American species. Furthermore, *Roccella verruculosa*, *R. bajasurensis* and *R. gracilis* were internally somewhat better resolved in the consensus tree.

Morphology

Thallus growth form with terete branches is present in *Roccella bajasurensis* and *R. verruculosa* as well as in all of the species in the *R. galapagoensis* group. Flattened, strap-like branches are present in two species, *Roccella decipiens* and *Roccella gracilis*. Besides their flattened branches they are also characterised by their yellow medulla within their holdfasts, however, that character is also shared by *R. bajasurensis*. The yellow pigmentation of the holdfast medulla is most prominent in *R. gracilis* and *R. bajasurensis*, although not as conspicuous as in the European–Macaronesian species *R. phycopsis*. *Roccella decipiens* usually shows a more dirty yellow colour resembling that of the European–Macaronesian species *R. tuberculata*, but sometimes this colour may be inconspicuous or even, but rarely, absent. *Roccella gracilis* is usually sorediate and the consistency of its thallus is relatively soft and it grows on the trunks and branches of various shrubs and trees but also, although not as frequently, on rock. *Roccella gracilis* is present with a fertile population in southern Baja California and neighbouring areas (Fig. 5) but fertile specimens have never been found growing on rock. *Roccella decipiens* has a much stiffer and rigid esorediate thallus, it is strictly saxicolous and it has a more northerly distribution (Fig. 3) than the allopatric fertile population of *R. gracilis*.

The *Roccella galapagoensis* aggregate contains five strictly saxicolous species of which four are endemic to the Galapagos Islands, and one is endemic to the Peruvian coast on the South American mainland. Two of the species, both restricted to Galapagos, are normally fertile, *R. albida* and *R. galapagoensis*. The other three species are typically sorediate: *R. margaritifera* and *R. nigerrima* are both found only in the Galapagos whereas *R. lirellina* is found only in Peru, on the South American mainland [see Tehler *et al.* (2009) for a detailed account on *R. galapagoensis* aggregate]. The only other *Roccella* species present in the Galapagos is *R. gracilis*. Small specimens of *Roccella gracilis* growing on rock could be easily confused with one of the sorediate species *R. margaritifera*, *R. nigerrima* or *R. albida*, but *R. gracilis* can be accurately distinguished by its C+ red soredia reaction and a yellow holdfast medulla. The other Galapagos *Roccella* species have invariably C-negative soredia and a non-yellow holdfast medulla.

The terete branched species, *Roccella bajasurensis* and *R. verruculosa*, are easily distinguished by the fact that *R. bajasurensis* has a yellow holdfast medulla and a C+ red thallus reaction whereas *R. verruculosa* is negative in those features. Infraspecific variation is moderate for all species with respect to morphology, anatomy and chemistry. Genetic variation that correlates to geographic distribution is notable in *Roccella*

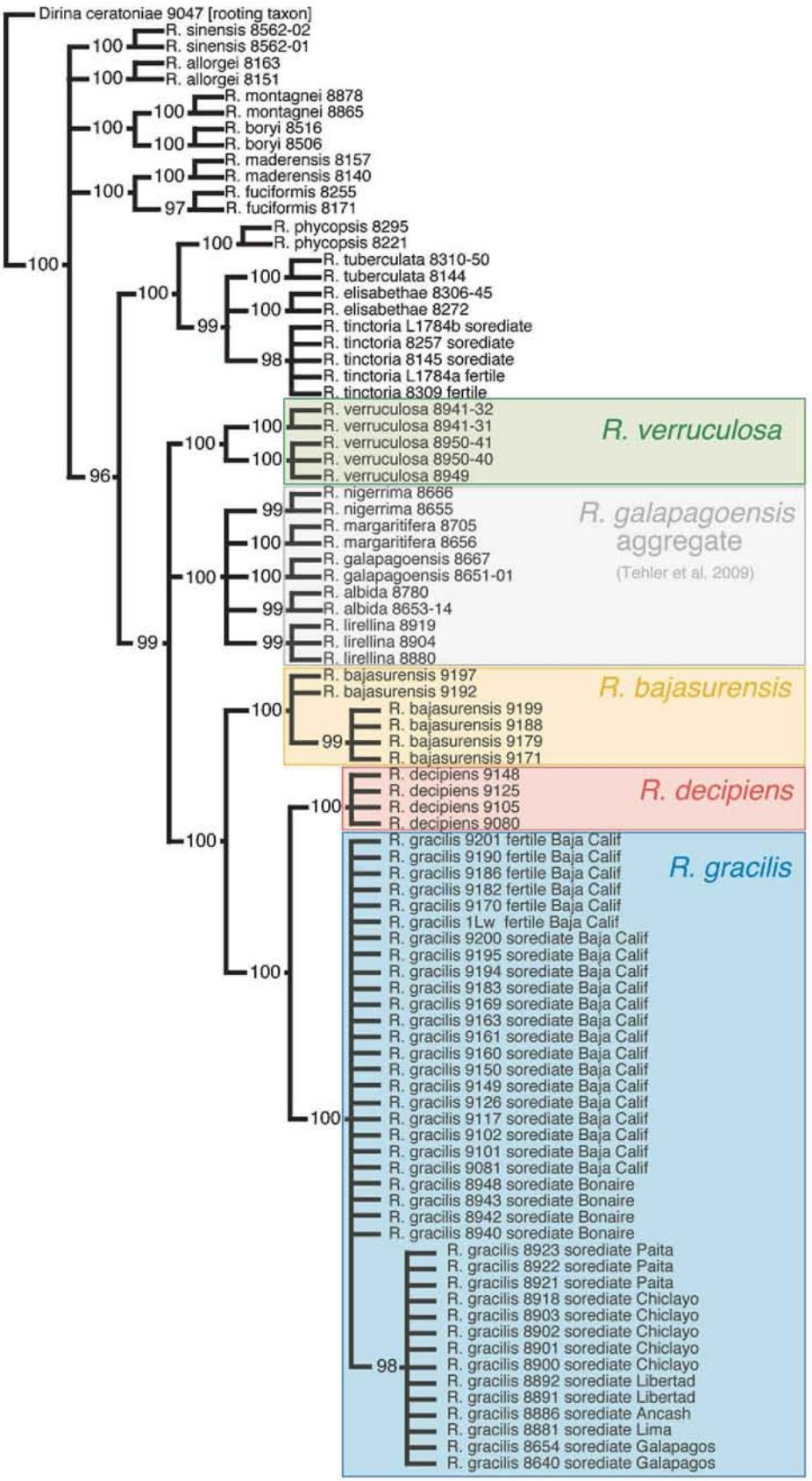

Figure 8 Parsimony jackknife tree received from the combined analysis of 89 sequences from RPB2, LSU, ITS1 and 2 and an anonymous locus. Jackknife frequency on node obtained from 1000 pseudoreplicates; only statistically significant nodes are shown. Taxon sampling complete from America and Macaronesia. A selection of three Old World species are included: *Roccella boryi*, *R. montagnei* and *R. sinensis*. Number after terminal name refers to collection number, see Table 1, which is available as 'Supplementary data' on Cambridge Journals Online: http://www.journals.cup.org/abstract_ S1477200009990065).

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gracilis (Figs 5, 8) with a sterile clone restricted to the Galapagos and Peru. *Roccella bajasurensis* also shows genetic variability, but no geographic correlation can be made because of its very limited distribution area at the tip of Baja California.

Discussion

Phylogeny

Probably the most surprising result of the present study was that the otherwise strictly sorediate *Roccella gracilis* included a strictly fertile population. Fertile specimens were earlier believed to belong to the species *Roccella decipiens*. *Roccella gracilis* and *Roccella decipiens* are morphologically and chemically similar having characteristically flattened, strap-like thallus branches and a yellowish medulla in the holdfast. *Roccella gracilis* was earlier believed to be a strictly sorediate species and possibly functioning as the sterile counterpart in a species pair relationship with the strictly fertile *R. decipiens*, which was considered as its fertile counterpart whether saxicolous or corticolous (Myllys *et al.*, 1999; Tehler, 2002). In this study we could show that saxicolous and corticolous fertile specimens belong to two distinct species respectively.

Sorediate specimens of *Roccella gracilis* are usually corticolous but saxicolous specimens also commonly occur across the entire distribution area although much less frequently. Similarly, it was believed that fertile specimens also used both corticolous and saxicolous habitats and they belonged to the species *R. decipiens*.

Molecular data from four loci convincingly indicate that strictly corticolous, fertile *Roccella* specimens from a population in Baja California Sur belonged to the otherwise predominantly sorediate species *Roccella gracilis*, and not to the closely related, fertile but strictly saxicolous species *R. decipiens*. Fertile, corticolous specimens are clustered in a significantly supported but largely unresolved group together with sorediate specimens of *R. gracilis* (Fig. 8). Corticolous, fertile specimens occur only in the southern half Baja California Sur and adjacent areas (Fig. 5). Equally convincingly molecular data indicate that strictly saxicolous, fertile specimens of *Roccella decipiens* form a taxon of its own (Fig. 8) having a more northerly distribution in the northern half of Baja California and southern California (Fig. 3).

Roccella gracilis as here circumscribed does not correspond to two distinct fertile and sorediate clades and there were neither molecular nor morphological evidence in the data that fertile and sterile, sorediate, specimens should be recognised as two distinct species (Fig. 8). Hence, evolutionarily they will be regarded as one common species, *Roccella gracilis*, as evident from the phylogenetic tree even if the two morphs are easily distinguished on the presence or absence of apothecia or soredia. Interestingly, transitional specimens that carry both apothecia and soredia are extremely rare. Exactly what mechanisms are responsible for the regulation of vegetative contra sexual reproduction modes are not known.

Roccella decipiens as here circumscribed is recovered in a sister pair relationship with the facultatively fertile/sorediate *R. gracilis* and the present results are yet another example in which the species pair concept again is falsified and shows that sorediosity cannot uncritically be used as a criterion for species delimitation (Lohtander *et al.*, 1998; Articus *et al.*, 2002).

The infraspecific variation may be conspicuous. *Roccella gracilis* is internally resolved into a significant group containing 13 sample sequences with a distribution restricted to the Galapagos Islands and the adjacent South American mainland in Peru (Fig. 8). All other 25 samples are contained in basal polytomy within the *Roccella gracilis* sample group with a more northerly distribution in the Caribbean (Bonaire) and upper and lower California (Figs. 5, 8). The polytomy includes 8 sequences from fertile specimens while the former Galapagos/Peruvian samples include only strictly sterile, sorediate specimens. As would be expected for a group with strictly vegetative dispersal the specimens are genetically very homogeneous with no significant internal resolution. We interpret the Galapagos/Peruvian group as originating from a sterile, sorediate and slightly different population that dispersed into the Galapagos region and successfully multiplied by vegetative means. It gave rise to the large, homogeneous population that now covers the geographic area with the Galapagos Islands and the coast of nearby Peru on the South American mainland.

The species thallus morphology with either terete or flattened branches are present also in species outside the study group and what character state is apomorphic or plesiomorphic when taking the whole genus into account remains to be investigated (Tehler *et al.* in preparation).

The resolution seen in the sample clades *Roccella verruculosa* and *R. bajasurensis* (Fig. 8) cannot be correlated to any morphological features and is interpreted as genetic variation within sexual populations similar to the situation reported from *Roccella galapagoensis* (Tehler *et al.*, 2009).

Biogeography

The American *Roccella* species were resolved into a disjunct, monophyletic group with nearly full support (99%) with the major part of the Macaronesian *Roccella* species supported by 100% as its sister group, and in turn both these groups together constitute a significantly supported group (96%). Outside the monophyletic American–Macaronesian group (Fig. 8) there is a number of other, plesiomorphic species all distributed in the Old World.

Disjunct geographic distribution patterns must be explained either by dispersal or by vicariance. The ancestor to the American–Macaronesian group may have dispersed from somewhere in the Old World, probably from the ancient Macaronesian region to the New World followed by directed distribution and adaptive radiation paralleled in both the Macaronesian and the American regions, resulting in the evolution of the two disjunct monophyletic American and Macaronesian species groups we see today. Both the American and the Macaronesian groups are significantly supported as monophyletic and it is obvious that the initial dispersal event leading to the evolution of the two groups has not been followed by any

successive long dispersal events that have had any impact by way of disturbing the distribution pattern.

For the alternative, vicariance, to be the case we see no other scenario than that through plate tectonics and continental drift with the expanding South Atlantic Ocean developing into a geographical dispersal barrier. The ancestor to the monophyletic American–Macaronesian species groups may have been distributed over an ancient geographical region that successively broke up into the New and Old World distribution areas. The American and Macaronesian populations became more and more isolated from one another during the late Jurassic 160–145 Mya and finally in the late Cretaceous 100–65 Mya became fully separated through the formation of the South Atlantic Ocean. Successively *Roccella* established and radiated into the present species in the two current distribution areas.

The hypothesis with vicariance determined by plate tectonics may seem somewhat inconceivable since it anticipates *Roccella* to be extremely old. The precursors to the genus must have evolved or had already evolved along the coasts of the western Tethys Sea and they must have had an established distribution at the opening of the Atlantic Ocean when Pangaea broke up into Laurasia and Gondwana in the Mesozoicum over 200 million years ago, thus, older than the first Angiosperms. On the one hand, this implies a very slow evolutionary rate compared with many other organisms, and also that easy cosmopolitan dispersal in lichens is not always the rule. On the other hand, despite the fact that conservative evolution would have prevailed for these organisms over times, dispersal and adaptive radiation have obviously also played an important role in the evolution of some species groups of *Roccella*, for example the rapid evolution of the *Roccella galapagoensis*-complex (Tehler *et al.*, 2009).

The most plesiomorphic species of the American Roccellas as suggested by the consensus tree (not shown) is *Roccella verruculosa* with the Galapagos species and the species that are endemic to the Californian region including the widespread *Roccella gracilis* placed as sister groups to one another (Fig. 8). The most likely infra-American distribution scenario, based on that hypothesis, although insignificantly supported, is that the *Roccella* migration in the Americas went forward in two directions. One route goes south into Peru and the Galapagos Islands, radiating into the *Roccella galapagoensis* group (Tehler *et al.*, 2009). The other route goes north to Upper and Lower California, radiating into *Roccella bajasurensis*, *R. decipiens* and *R. gracilis*. The latter species, which is fertile only in Baja California and strictly sorediate in all other areas, has obviously dispersed back to the Caribbean and further south to Peru and the Galapagos Islands.

Since the species of *Roccella* are mainly restricted to maritime coastal habitats they consequently often prefer island substrates such as those found in, for example, the Galapagos Islands, the Canary Islands, the Azores, Socotra and Mauritius, all islands from which the age can be accurately determined and thus could serve as calibration points for the nodes in a phylogenetic tree. This situation will be investigated in more detail in a forthcoming publication when all the species in *Roccella* have been thoroughly studied and circumscribed.

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